Example 11.2

Consider the AR(1) process

$$X_t = \phi X_{t-1} + Z_t$$

(assume that the process is (weakly) stationary)

$$\Rightarrow (1-\phi B) X_t = Z_t$$

$$\Rightarrow X_t = (1-\phi B)^{-1} Z_t$$

$$= \sum_{j=0}^{\infty} \phi^j Z_{t-j}$$

(Taylor series expansion of $(1-\phi B)^{-1}$)

$$= \sum_{j=0}^{\infty} b_j Z_{t-j}$$

where $b_j = \phi^j$.

We know that

$$\hat{X}_{n+k}(n) = \sum_{u=0}^{\infty} b_{u+k} Z_{n-u}$$

$$= \sum_{u=0}^{\infty} \phi^{u+k} Z_{n-u}$$

$$= \phi^k \left[\sum_{u=0}^{\infty} \phi^u Z_{n-u} \right]$$

$$\sum_{u=0}^{\infty} b_u Z_{n-u} = X_n$$

$$= \phi^k X_n.$$